Introduction (undergraduate level)

The spatial dimension of development: The gravity equation

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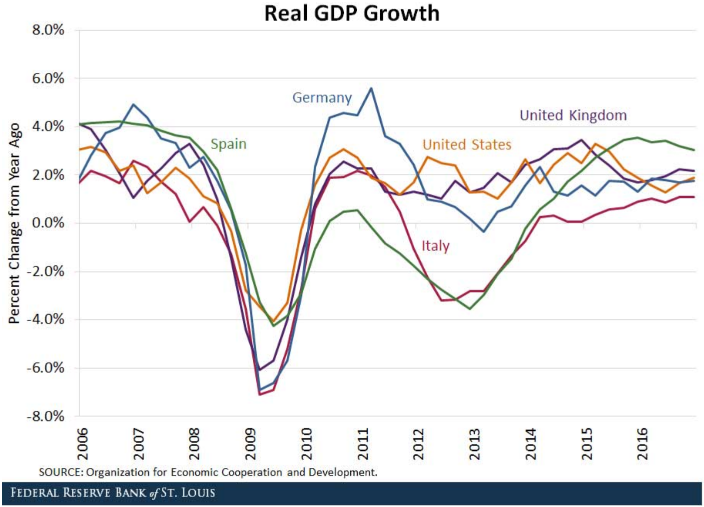
Introduction

- The economic development of countries is interconnected.



- One channel for this connection is trade.

Why study trade?

Trade is a centuries-long topic in economics.

- The first formal economic model was about trade, written by the Scottish philosopher David Hume in "Of the Balance of Trade".
- In 19th century Britain, there was a heated debate about whether to promote free trade (the Corn Laws).
- David Ricardo formalised the idea of comparative advantage in support of free trade.
- The Corn Laws were repealed in 1846 after Ricardo's death. His free trade paradigm became public policy.

Why study trade?

- Trade has been a reality for a long time.

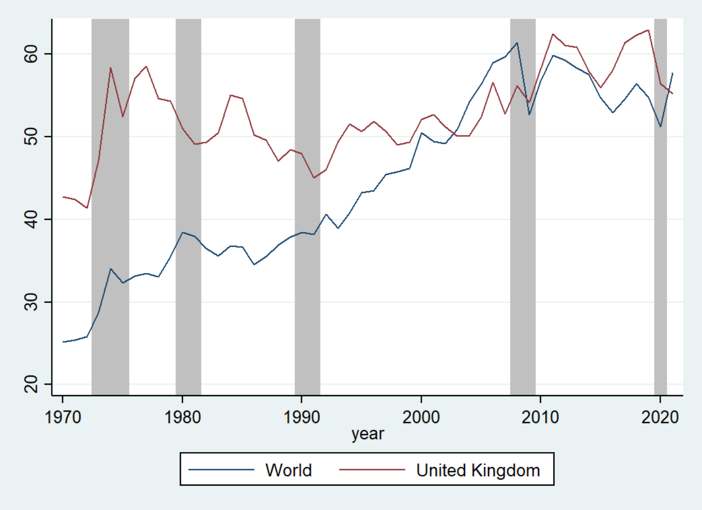
Cuneiform tablet: caravan account ca. 20th–19th century B.C. Kültepe.

Source: Spar (1988)

- long distance travel by Assyrian merchants between Mesopotamia and Anatolia (6-8 weeks)
- trade tin and textiles for gold and silver
- Mesopotamian tools used to record transactions: cuneiform writings in clay tablets
- This text records investments in trade items (tin and textiles), expenditures such as clothing and wages for guides.

Why study trade?

- Trade has been a reality for a long time and has become increasingly important.



Trade as percentage of GDP (shaded areas indicate UK recessions)

Why study trade?

- Gains from trade
  - Trade benefits all countries.
  - Reason 1: comparative advantage (Lectures 2&3)
  - Reason 2: inequality in endowment (ECNM10087)
  - Reason 3: specialisation and economy of scale (ECNM10087)
- Inequality from trade (Lectures 4&5)
  - Within a country, there are winners and losers from trade.
  - Free trade hurts owners of immobile factors of production.

Plans

- Today: Patterns of trade (the gravity equation)
  - Trade between two countries is proportional to their economic sizes and inversely proportional to the distance between them.
- Weeks 2 and 3: Comparative advantage (the Ricardian model)
  - The simplest trade model
  - 2 countries, 2 goods and 1 factor of production
  - Countries produce relatively cheaper in one good (comparative advantage).
  - This generates gains from trade.
- Weeks 4 and 5: Inequality of trade (the specific factors model)
  - 2 countries, 2 goods and 2 factors of production
  - One factor (specific) is not mobile.
  - This produces winners and losers from trade.

Methodologies

- Today we look at general patterns and determinants of trade.
- From the next week onwards, we will introduce (general equilibrium) trade models.
- We will combine economic intuitions, graphical analysis and math.
  - in lectures and tutorials.

Why theories and models?

- It is intuitive to see that trade decreases with distance.
- However, some phenomena are (more) difficult to explain.
- For instance, to answer the question why countries trade we first come up with a hypothesis.
  - e.g., Comparative advantage.
- To test whether the hypothesis is correct, we build a model assuming it is true.
- The equilibrium of the model tells us whether going from comparative advantage we reach an outcome where countries trade.
- Models also make possible counterfactual analysis and hence welfare comparison.
  - e.g., We can compare model outcomes in autarky and free trade to see whether countries are better off with trade.

Why theories and models?

- Caveats: Models are not perfect.
- Models are simplified versions of the real world.
- By isolating the unimportant, we are able to examine a particular mechanism (e.g., Comparative advantage).
- We start with clear assumptions and follow the model to its logical conclusion.
- Therefore, it is helpful to think about
  - how these assumptions affect the results of the model.
  - what aspects of the real world are missing in the model and whether they will change the outcomes (to a lesser or a greater extent).
- It is useful to build a simple model first and add more things later.

Patterns of trade

- We begin by looking at patterns of trade:
  - Some countries trade more than other countries.
  - Country A has more trade with Country B than with Country C.
  - ...
- We can use the **gravity equation** to characterise trade patterns and analyse the determinants of trade.
- It was first introduced by Jan Tinbergen (the first Nobel Prize winner in Economics) in 1962.
  - It was not theory-based at the time but is now micro-founded by trade models.
  - It says that bilateral trade flows between any two countries are proportional to the economic sizes of those countries and inversely proportional to the distance between them.

The gravity equation

The gravity equation is:

$$T_{ij} = C \times \frac{Y_i \times Y_j}{D_{ij}}. \tag{1}$$

- T_{ij} is the value of trade between country i and country j.
- C is a constant.
- Y_i and Y_j are GDPs of country i and country j respectively.
- D_{ij} is the distance between the two countries.

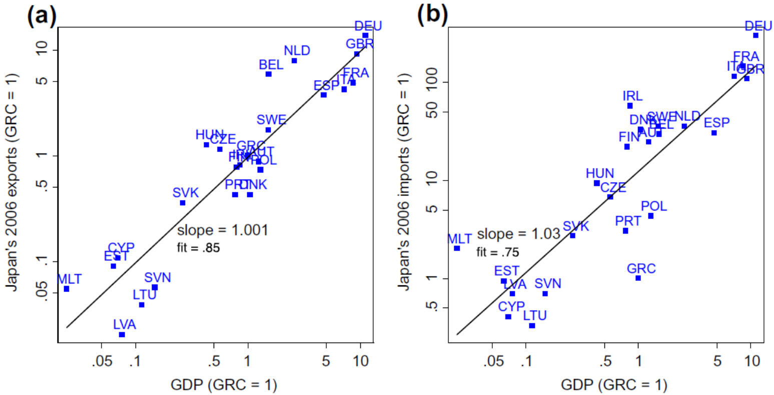
What does the equation tell us?

- Trade between any two countries increases with (and is proportional to) their GDPs.
- Bilateral trade diminishes with (and is inversely proportional to) distance.

Does this make sense?

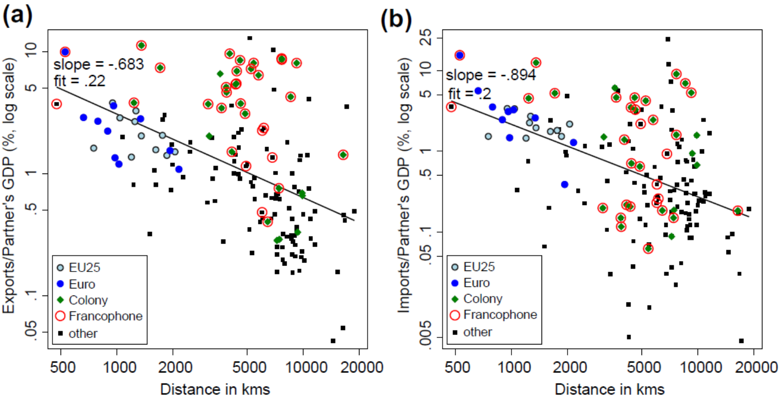
Note: the name of the model comes from the analogy to Newton's Law of Universal Gravitation. Paul Krugman refers to it as an example of "social physics."

Does gravity match the data?



Source: Head and Mayer (2014)

Does gravity match the data?



Source: Head and Mayer (2014)

Note: We can rewrite the gravity equation as $\frac{T_{ij}}{Y_j} = CY_i \times \frac{1}{D_{ij}}$. After taking natural logs on both sides, we have $\ln \frac{T_{ij}}{Y_j} = \ln(CY_i) - \ln D_{ij}$.

The gravity equation

A more general version of the gravity equation is:

$$T_{ij} = C \times \frac{Y_i^{\alpha} \times Y_j^{\beta}}{D_{ij}^{\theta}}. \tag{2}$$

By taking natural logs on both sides, we have:

$$\ln T_{ij} = c + \alpha \ln Y_i + \beta \ln Y_j - \theta \ln D_{ij}, \tag{3}$$

where $c = \ln C$. After adding an error term ϵ_{ij}, we can estimate the equation using data:

$$\ln T_{ij} = c + \alpha \ln Y_i + \beta \ln Y_j - \theta \ln D_{ij} + \epsilon_{ij}. \tag{4}$$

Log-log coefficients have straightforward interpretations:

- For example, a percentage decrease in distance is associated with a θ percentage increase in trade.
- We refer to them as the elasticities of trade.
- The estimated α and β are close to 1, indicating that trade is proportional to economic sizes.
- Economists are mostly interested in $-\theta$: the distance elasticity of trade.

Impediments to trade

In general, in addition to distance, we can include other measures of bilateral resistance. For example:

- tariffs (+resistance),
- free trade agreements (-resistance),
- common borders, language and history (-resistance).

Impediment to trade: Distance

The distance elasticity is stable over time and across regions.

- Using data on trade flows between countries from 1950 to 2000, the distance elasticities are estimated to be between -1.5 and -0.5 (Allen and Arkolakis, 2018).
- The distance elasticity is -1 for trade between states in the U.S (Allen and Arkolakis, 2018).
- Trade in terra sigillata during the Roman Empire has a distance elasticity of -1.5 (Flückiger, Hornung, Larch, Ludwig, and Mees, 2021).

- Meta-analysis of Head and Mayer (2014): The average distance elasticity is -0.93 (standard deviation: 0.4) in 159 papers (more than 2500 estimates).

Impediment to trade: Distance

Why distance can be used to measure impediments to trade?

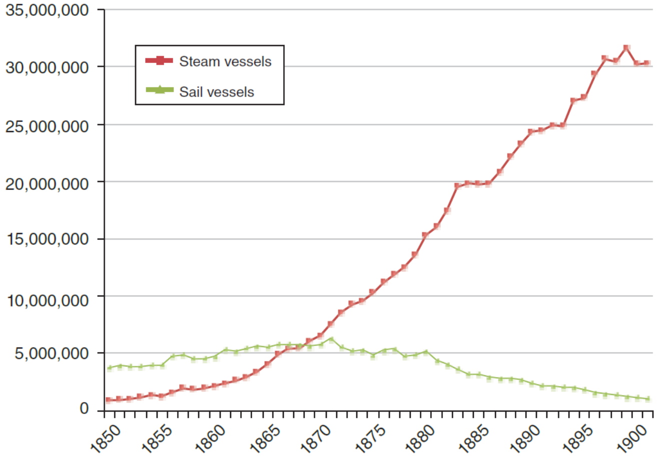
Distance is a proxy for

- transportation costs,
- iceberg trade costs,
- communication costs,
- transaction costs.

Impediment to trade: Transportation costs

- Transportation costs can change due to technological change or infrastructure investment.
- Pascali (2017) looks at how the introduction of the steamship in the 19th century changed world patterns of trade.
- Steamships were a technological improvement over the incumbent: Sailing ships.
  - Sailing depends heavily on wind.
  - Steamships experienced major technological breakthroughs in the 1850s and 1860s.

Impediment to trade: Transportation costs



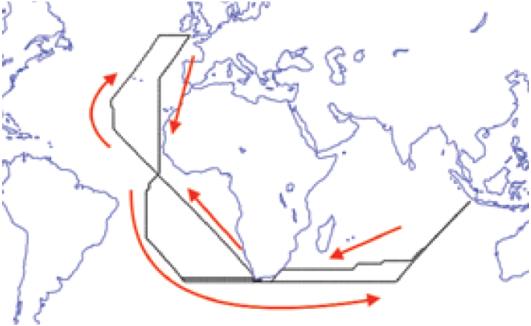
Total Tonnage of British Vessels Entered in British Ports from and to Foreign Countries and British Possessions

Source: Pascali (2017)

Impediment to trade: Transportation costs

The advance in sea transport technology induced an asymmetric change in shipping time across trade routes.

- E.g., The duration of a round trip for a sailing vessel from Lisbon to Cape Verde would be similar to that of a round trip from Lisbon to Salvador (due to the clockwise wind pattern in the Northern Atlantic Ocean).
- With the steamship, the duration of the former trip would be only half of the latter.



Optimal routes by sail

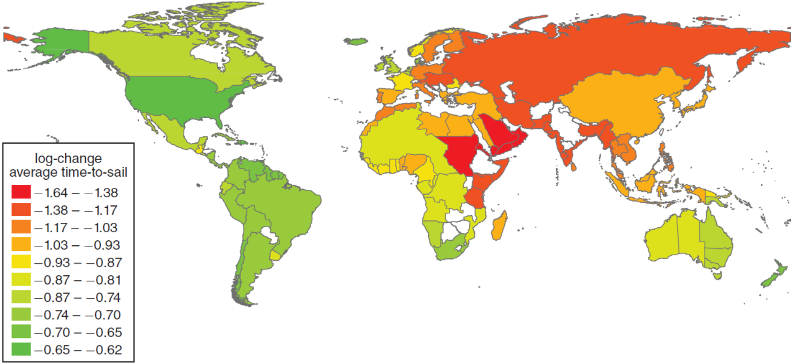
Impediment to trade: Transportation costs

- All locations benefited from the technological change, but some benefited more than the others.
- The author calculates the change in travel time due to the introduction of the steamship for each trade route.
- Then aggregate to the country level to calculate a measure of transportation cost reduction for each country:

$$\Delta \ln Dist_i \equiv \sum_{j \neq i} w_j [\ln(steamTIME_{ij}) - \ln(sailTIME_{ij})], \qquad (5)$$

where weight w_j is country j's share in world trade.

Impediment to trade: Transportation costs



Log-change in the average travel time (from sail to steam)

Source: Pascali (2017)

Impediment to trade: Transportation costs

Then examine whether the reduction in shipping time is associated with an increase in trade by testing the following equation:

$$\Delta \ln T_i = c - \theta \Delta \ln Dist_i + v_i, \tag{6}$$

where T_i is either the export-to-GDP ratio or the per capita export, c is a constant, and v_i is an error term.

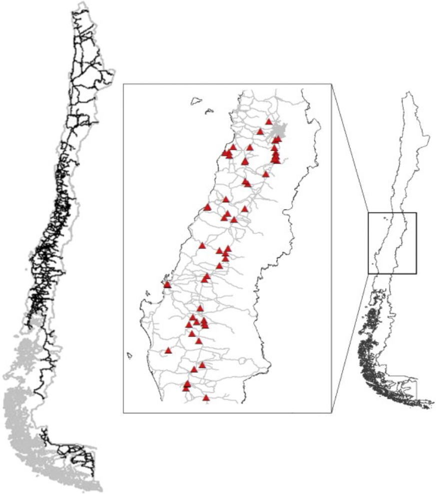
The paper finds that:

- 1 percentage reduction in travel time is associated with 1 percentage increase in the export-to-GDP ratio and 1.7 percentage increase in the per capita export.
- The introduction of the steamship can account for about 50% of the trade boom that occurred over this period (i.e., the first wave of globalisation).

Impediment to trade: Transportation costs

- Transportation costs can also change because of infrastructure investment.
- If infrastructure improves, we expect transportation costs to drop $\rightarrow$ trade increases.
- Conversely, destruction of transportation infrastructure can reduce trade.
- Martincus and Blyde (2013) look at a negative shock to the transportation infrastructure in Chile and investigate its effect on export to foreign countries.
- On February 27, 2010, an earthquake with a magnitude of 8.8 took place in Chile.
- The earthquake caused severe damage to the transportation network (9% paved road network).
- Repair took several months.

Impediment to trade: Transportation costs

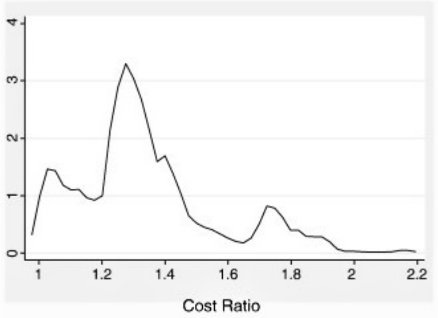


Chile's primary and secondary road network and the traffic interruption points (red triangles).

Source: Martincus and Blyde (2013)

Impediment to trade: Transportation costs

- Presumably, firms had used the least-cost-routes from the production facilities to the existing custom points for foreign export.
- Some of the routes became impassable due to the destruction caused by the earthquake.
- As a result, some firms, but not others, had to use alternative routes, causing an increase in the transportation costs.



Distribution of (new cost/old cost)

Source: Martincus and Blyde (2013)

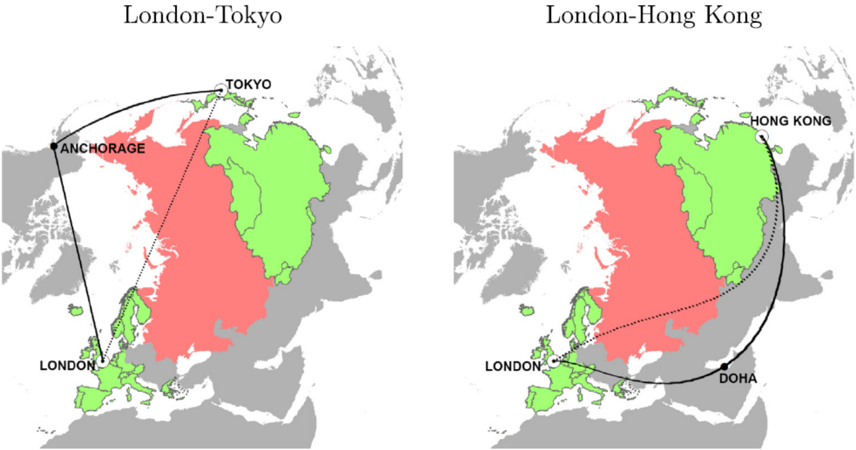
Impedment to trade: Transportation costs

- The authors find that if the export's routes were damaged by the earthquake and had to be rerouted, trade reduced by 34%.
- They control for the direct effect on production disruption.
- Without changes in domestic road infrastructure, total industrial exports would have been 6.3% larger over the period of February 27, 2010 to February 26, 2011.
- The distance elasticity of trade is found to be -1.42.

Impediment to trade: Communication costs

- According to Krugman, distance "proxies for the possibilities of personal contact between managers, customers, and so on; that much business depends on the ability to exchange more information, of a less formal kind, than can be sent over a wire."
  - Campante and Yanagizawa-Drott (2018) find that air links increase business links.
  - Söderlund (2023) uses the liberalisation of the Soviet airspace to identify the effect of reduction in business travel time on trade.

Impediment to trade: Communication costs



Change in flight routes before and after the 1985 liberalisation

Source: Söderlund (2023)

Impediment to trade: Communication costs

- Söderlund (2023) finds that the liberalisation led to a substantial increase in (non-stop) passenger traffic for affected country pairs.
- The estimated effect on trade is also large.
  - The estimated distance elasticity is -1.17.
  - This means that a 10% reduction in flight time is associated with an approximately 11.7% increase in goods trade.
- The effect is not driven by a lower transport cost (using goods not transported by air).
  - Business travels $\rightarrow$ face-to-face interactions $\rightarrow$ business relationships $\rightarrow$ trade.
- Business travel time/communication costs can account for a significant share of the distance elasticity of trade.

Impediment to trade: Borders

- The Canadian-U.S. border is one of the most open borders in the world.
- The two countries are part of a free trade agreement (previously NATFA, now USMCA).
  - Most goods shipped across the national border pay no tariffs.
- They share a common language.
- British Columbia has 9.5 times more trade with Ontario (in Canada) than with Ohio (in the U.S.).

Is this because of distance?

Trade with British Columbia, as Percent of GDP, 2009			
Canadian Province	Trade as Percent of GDP	Trade as Percent of GDP	U.S. State at Similar Distance
Alberta	6.9	2.6	Washington
Saskatchewan	2.4	1	Montana
Manitoba	2	0.3	California
Ontario	1.9	0.2	Ohio
Quebec	1.4	0.1	New York
New Brunswick	2.3	0.2	Maine

Source: Statistics Canada, U.S. Department of Commerce.

Impediment to trade: Borders

- High border effects exist for European trade as well.
- Why do borders make a difference after controlling for distance?
  - National institutions vary across borders (e.g., legal, monetary, social, cultural).
  - history and established business relations

Impediment to trade: Institutions, history and culture

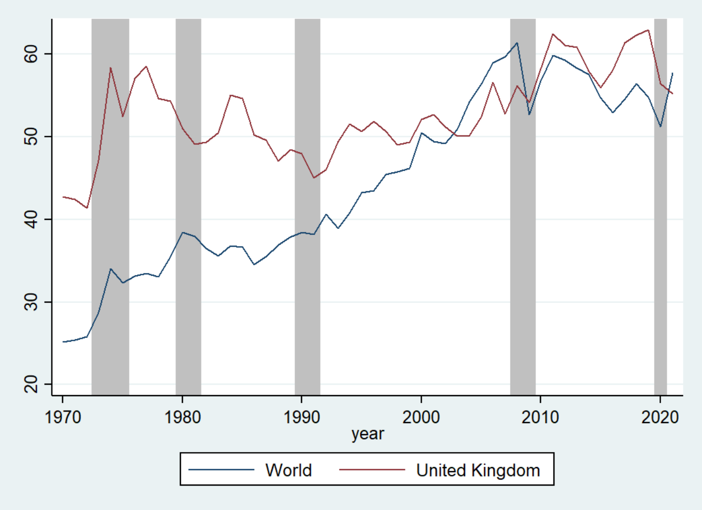
- Common currency:
  Two countries that share a common currency trade twice as much as country pairs that do not.
- Common language:
  Two countries that share a common language trade twice as much as country pairs that do not speak the same language.
- Colonial links:
  Countries with colonial links trade 2.5 times as much as country pairs without colonial links.

The gravity equation and policy evaluation

- The gravity equation can be used to evaluate the effect of regional trade agreements or free trade agreements.
  - Trade agreements can be thought of as the opposite of impediments to trade.
- As expected, these agreements reduce bilateral resistance and can increase trade.
  - The North American Free Trade Agreement (NAFTA) is associated with a 50% increase in bilateral trade.
  - The European Union is associated with a 15% increase in bilateral trade.

The gravity equation and patterns of trade

The gravity equation can explain the general increase in trade and its fluctuations.



Trade as percentage of GDP (shaded areas indicate UK recessions)

Summary

- Trade and development are interconnected.
  - How much countries trade is related to their economic sizes.
- Improvements in transportation and communication technologies and the reduction of trade barriers (e.g., free trade agreements) have contributed to significant growth in trade.
- The distance elasticity of trade is stable over time and across regions.

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